



Age-related disease or disease-related age? Perspectives for paleopathological research

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ABSTRACT

Objective: The interconnection of biological aging and disease throughout the life course profoundly affects almost every aspect of paleopathological research but is rarely explored. This paper presents a theoretical and practical framework for an informed research approach that explicitly acknowledges these complex interrelationships.

Methods: We discuss the existing concept of *age-related disease* and introduce the new concept of *disease-related age*. Their relevance and practical applicability are demonstrated through selected paleopathological examples, which propose modifications to existing practice and avenues for future research.

Conclusion: Diseases and pathological lesions influence the ability to estimate skeletal age at death, whilst our inability to accurately estimate age and death affects our ability to understand past disease. By incorporating consideration of *age-related disease* and *disease-related age* into paleopathological research, alongside adopting specific methodological approaches, we will significantly improve our understanding of both past disease and past demography.

Significance: We offer the first comprehensive assessment of the nature and implications of skeletal age-disease interrelationships and their impact on researching past health and disease.

Limitations: This paper is the first step forward in engaging with the complex, but critically important, issues of the inter-relationship of age and disease in paleopathology.

Suggestions for further research: The nature of the interaction between pathophysiology and aging processes and its osseous manifestations require further interdisciplinary research, both on a case-specific and systemic level. Results will need to be integrated into osteobiographical, paleodemographical, and paleoepidemiological studies.

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1. Introduction

Since the founding of the Paleopathological Association 50 years ago, palaeopathology has established itself as a formal scientific field (Grauer, 2018). Paleopathologists routinely deal with the complex challenges of assessing hard and preserved soft tissue abnormalities, exploring molecular evidence of health and disease, developing and interpreting differential diagnoses, estimating individual disease histories, and reconstructing paleoepidemiological patterns, all whilst incorporating social and evolutionary theory, and holistic and biocultural conceptions of health and disease (Grauer, 2018; Grauer and Buikstra, 2019). One important topic which remains largely unaddressed, though, is the interplay between age and disease. Yet, aging and disease are inherently connected. This interconnection affects almost every aspect of paleopathology, including our understanding and interpretation of bone biology, our methodological practice, our ability to reconstruct the relationship between bony changes and functionality, and our conceptualization and interpretation of age and aging.

Both osteological age-at-death estimation and paleopathological inquiry are significantly affected by the fact that there is no simple relationship between chronological and biological age. Whilst chronological age refers to age [since birth] in years, biological age relates to processes of physiological development, bodily maintenance and degeneration, and is *not* simply a matter of straightforward change over time. Studies have revealed non-linear and sometimes considerable discrepancies between chronological and phenotypic age in living humans and chronological and skeletal age in osteological collections (e.g., Couoh, 2017; Pala et al., 2024). Biological aging is complex and interlinked with genetic, environmental, economic, cultural, and social factors. Associations between estimated age, indicators of health, disease, stress, and frailty, and aspects of social identity often remain unclear, whilst disease and aging are themselves intertwined to such an extent that we will probably never be able to fully disentangle them.

Aging, therefore, has two interrelated effects on paleopathological inquiry. On one hand, many diseases are either associated with specific age groups or change in prevalence and severity with age. On the other hand, disease processes may interact with bodily processes to affect the age markers used to establish age at death. Thus, diseases affect our ability to accurately estimate age at death, whilst our inability to accurately estimate age at death affects our ability to understand past diseases. Although the inherently close relationship between age, aging, physiological stress, disease and its nature of cumulative exposure and chronicity is commonly known (e.g., Mays, 2015; Milner and Boldsen, 2017; van Schaik et al., 2018), research on this topic is limited within paleopathology and has yet to become an integral part of the field.

This perspective paper presents some strategies for how to approach the relationship between age and disease in paleopathological practice. We use two concepts—*age-related disease* and *disease-related age*—as useful tools for outlining the interactions between disease and the aging process, as well as estimating their influence on skeletal marker morphology. Using examples, we demonstrate the relevance and application of these concepts so that they can be considered more effectively in the field's future. In response to the seemingly intractable issue of the inter-relationship of age and disease, we provide recommendations for adjusting our everyday practice and suggest avenues of research that could lead the way forward.

2. Age-related disease or disease-related age?

2.1. Age-related disease

Age-related disease, while a frequently used term with an apparently common-sense definition, has no universally-accepted definition in the clinical or public health literature (Appleby, 2024; Le Couteur and Thillainadesan, 2022). Until now, paleopathological inquiry has proceeded on the assumption that “normal” aging and disease can, to at

least some extent, be separated out for the purposes of analysis; however, close reading of the literature on *age-related disease* reveals that there is no clear consensus on how to identify and separate diseases that are inherently caused by the aging process itself (sometimes distinguished as aging-related diseases) and those that are merely associated with age due to (for example) cumulative exposure to risk factors (age-related; Kuan et al., 2021; Le Couteur and Thillainadesan, 2022). Some diseases that are commonly cited as diseases of aging are not age-progressive in all cultural contexts. For example, amongst the Tsimane in Bolivia, low exposure to risk factors means that cardiovascular disease risk does not increase with age (Strandberg and Kivimäki, 2022), despite the normal assumption that cardiovascular disease is an aging-related disease. This leaves the boundary between normal degenerative changes of aging and *age-related disease* unclear.

It is worth noting that the term *degenerative* refers to time-related breakdown of cells or structures and can be associated with the idea of aging-related diseases, but it is often used interchangeably with *degeneration*, which is loss of specialized cellular function due to defect or disease and not necessarily caused by aging (Wigley et al., 2011; “degeneration”, Oxford Reference, 2025). This terminological discussion underscores the paleopathological need to address the issue of which diseases are truly age-progressive in a given cultural context, rather than simply assuming age-related progression.

Our understanding of *age-related disease* is also hampered by our definitions of both aging and disease. Instinctive understandings of aging belie the fact that there is little consensus amongst gerontologists about what aging actually is (Chmielewski, 2020; de Magalhães, 2024). In addition, the definition of the term “disease” has been subject to considerable discussion (e.g., Fisher et al., 2025; Moynihan et al., 2013). A common approach is to define it as a “harmful” deviation from the norm (e.g. Burrows and Scarpelli, 2024; Funkhouser, 2020). This definition cannot apply to *age-related disease* since, as noted, many such diseases, e.g., various forms of cardiovascular disease, can be seen as normal within a given context. *Age-related diseases* are certainly harmful, but so are many of the processes of “normal” aging (Appleby, 2024; Caruso et al., 2021; see O'Rourke and Ceci, 2013). The difficulty in locating the boundary between aging and disease is thus partly due to difficulties in defining aging and disease, partly due to the lack of clarity in categorization, and partly because we do not have a clear way to separate *age-related disease* from other pathological processes that an individual experiences over the life course.

2.2. Disease-related age

It is commonly accepted that disease progression may be linked to age, and this is something that is frequently discussed in paleopathological studies. Yet the corollary of this—that skeletal age can be affected by disease—is rarely explicitly considered. In fact, many disease processes can influence skeletal age markers and thus our final age estimates, making skeletal age a *disease-related age*. In principle, any skeletal age indicator can be exposed to disease processes at any stage of life. However, this circumstance may not be reflected in either reporting or the way the age estimates obtained are used, such as in a biological profile. In light of the importance of age at death as an analytical parameter, this omission is worrying.

Whilst it is easier to ascertain a precise and accurate age at death for the skeletally immature than for adults, even in juveniles most age markers can be affected by chronic diseases. For example, premature epiphyseal fusion is associated with vitamin C deficiency (scurvy), hypervitaminosis A, certain infectious diseases such as tuberculosis, and various forms of anemia (Currarino and Erlandson, 1964; Lawson et al., 1983; Parke et al., 1949). Similarly, in both clinical medicine and paleopathology, long bone length is simultaneously used as both a criterion for age-typical growth and a signifier of various developmental disorders, such as malnutrition or other types of physiological stress (e.g., stunting) (Decrausaz and Cameron, 2022). In adults, obtaining

narrow age estimation ranges is a serious methodological issue. Mays (2015) notes that for a given skeletal assemblage, approximately 60 % of variation in adult skeletal age markers can be caused by factors other than age. Perhaps the most notable disease process involved in this variation is osteoarthritis (e.g., Calce et al., 2018; and see Section 3.4). On a more systemic scale, other factors such as vitamin D deficiency and concentrations of circulating sex steroid hormones are also influential, which may have implications for bone remodeling in individuals with these diseases. In fact, any condition affecting bone remodeling or cartilage ossification may also affect at least some age estimation methods (Mays, 2015). While the multifaceted aetiologies of the morphology of entheses, crucial for some aging methods such as Transition Analysis 3 (TA3, Milner et al., 2019) are recognized, the influence of pathogenic factors (e.g., collagen malformation, ruptures, inflammation) is typically difficult to grasp and operationalize in an age-related context.

Despite significant levels of interest in the reasons for variation in skeletal age markers, there has been very little published investigation into how disease contributes to skeletal aging beyond consideration of osteoporosis, joint and dental disease. In a recent content analysis of the literature on age estimation methods (Campanacho and Alves-Cardoso, 2023) the words “disease” and “pathology” are not mentioned. This is a significant research lacuna.

A fundamental problem for this endeavor is the lack of independent skeletal data on chronological age for disentangling, or “calibrating”, the biological processes of aging and the pathological processes of diseases. Even in living populations, there is no biomarker for age that may be unaffected by pathological processes (Marcante et al., 2025). Osteological age estimation in adults is both imprecise and inaccurate. This is because of a number of methodological and statistical problems, which have been discussed extensively over the years (e.g., Bocquet-Appel and Masset, 1982; Konigsberg and Frankenberg, 1992, 1994; Boldsen et al., 2022; Buckberry, 2015; Mays, 2015; Hoppa and Vaupel, 2002;). Statistical methods designed to overcome these issues (e.g., Konigsberg and Frankenberg, 1992, 1994; Moser et al., 2024; Müller-Scheeßel et al., 2025; McFadden et al., 2019; Séguin and Buchet, 2013; Rizos et al., 2024) can be effective for reconstructing age structure of past populations, but they cannot give a precise and accurate age or age range for individual skeletons. This also applies to the technique of counting tooth cementum annulation, whose potential as a disease-independent chronological age marker is under discussion (Bertrand et al., 2016, 2019; Couoh, 2017; Foster et al., 2022; Meinel et al., 2008).

These issues make it difficult to disentangle the relationship between chronological age and the development of particular diseases, which in turn has a significant impact on our ability to understand which diseases progressed with age in the past and how.

3. Age-related disease and disease-related age in current paleopathological research

3.1. Frailty and resilience

While the concept of frailty was introduced to the field of demography by Vaupel and coauthors in 1979, it was only with Wood and colleagues' (Wood et al., 1992) presentation of the osteological paradox that it became widely discussed in paleopathology. The concept of frailty has been operationalized in many ways in paleopathology, but perhaps most commonly in calculating the risk of mortality or increased hazard of death of an individual relative to other individuals within a population (Vaupel et al., 1979). Many paleopathological and bioarchaeological studies have effectively applied this concept of frailty in relation to mortality risk, with consideration of hidden heterogeneity in frailty and selective mortality in past populations (e.g., DeWitte, 2010; DeWitte and Stojanowski, 2015; DeWitte and Wood, 2008; Wyatt et al., 2022; Yauussy et al., 2016, 2024). Paleopathologists have also evaluated how specific lesions impacted frailty in past populations (Biehler-Gomez

et al., 2025a; DeWitte, 2014). However, while it is widely accepted that pathological skeletal lesions do not incur equal frailty across once-living populations or over time, the calculation of frailty is inherently linked to the estimated age-at-death, making it difficult to disentangle the interpretation of lesions from the aging process.

In gerontology and clinical medicine, frailty is a concept that encompasses limitations and declines in functions and capabilities (Fried et al., 2001; Walston et al., 2006). As a phenotype that reflects the accumulation of a wide variety of physiological and pathophysiological processes, frailty typically increases into and during older age (Crews, 2022; Rockwood and Mitnitski, 2011). This means that understanding frailty in its clinical sense is critical to understanding the aging process both in individuals and populations in past societies (Crews and Marklein, 2023; Marklein and Crews, 2017; Marklein and Crews, 2022; Petrosino et al., 2025; Tuggle et al., 2021; Zedda et al., 2021).

In studies of living populations, we can statistically evaluate how frailty changes with chronological age (Rockwood et al., 2004; Romero-Ortuno and Kenny, 2012). In the skeleton, however, estimated skeletal age at death and frailty may result from the same or similar processes, and it is oftentimes difficult, if not impossible, to systematically separate the two (Marklein et al., 2016; Marklein and Crews, 2022). In turn, this means that it can be difficult or impossible to empirically compare frailty between past populations with respect to specific patterns of aging, mortality, and disease. In one past population, *age-related diseases* may have appeared earlier (or later) in life than in another; the aging process itself may manifest differently on the skeleton; or markers may have been differentially affected by pathological processes. For example, a recent diachronic study comparing individuals from five historic contexts (Roman to Contemporary periods) in Milan has demonstrated how frailty varied by lesion presence amongst individuals between periods (Biehler-Gomez et al., 2025b). However, site- or period-specific frailty indices may provide ways to identify patterns in age-related diseases within certain archaeological contexts (Yauussy et al., 2024).

The corollary of the relationship between frailty, aging and disease is that, what at first glance looks like a difference in the chronological age distribution of frailty indicators may not hold true. This has significant implications for the use of concepts of frailty in paleoepidemiology and paleodemography.

3.2. Aging and inflammation

One major factor affecting frailty can be inflammaging. Researchers in gerontology and immunology have long linked chronic, low-level inflammation to increased age, otherwise known as inflammaging (e.g., López-Otín et al., 2023a). This increased chronic inflammation features systemic manifestations as well as localized phenotypes (e.g., intervertebral disc disease) (Baechle et al., 2023; Sanada et al., 2018). Dysregulation of the expression of signaling immune proteins, or cytokines, likely plays a crucial role in reshaping the immune system at an older age, with evidence pointing to an aging body's increasing inability to fine-control systemic inflammation (Rea et al., 2018). Immune function declines alongside increased inflammation (Carrasco et al., 2022), leading to systemic chronic inflammation or a systemic inflammatory phenotype. Inflammaging may also occur as a result of other systemic and localized hallmarks of aging (e.g., epigenetic alterations, telomere attrition; López-Otín et al., 2023a), and it is influenced by different biological, dietary, social, and environmental factors that affect inflammatory responses and overall systemic inflammatory phenotypes. This means that chronic, low-level inflammation, like other immunological processes (Crespo and Lawrenz, 2014; McDade, 2005), may be highly contextualized and manifest differently in past spatio-temporal and cultural contexts. Indeed, recent research in human biology and immunology has demonstrated that inflammaging is not a universal hallmark of aging but is linked to extrinsic factors (Frank et al., 2025a, 2025b).

The understanding that increased chronological age can be associated with low-level, chronic inflammation and systemic inflammatory phenotypes will likely help refine paleopathological work on the relationship between aging and disease in older adults. Future research could explore how biological age could influence the systemic dysregulation of immune responses and vice versa, while attending to the non-universality of inflammaging and the potential for substantial variation between cultural and spatiotemporal contexts. A chronic dysregulation of inflammatory responses could accelerate aspects of biological aging, accounting for mismatches between skeletal and chronological age. Inflammaging and a systemic inflammatory phenotype are likely to be underlying, silent players in other aspects of aging for some past populations, though potentially less so in pre-industrial contexts. Chronic inflammatory changes associated with aging also contribute to reduced bone density (see [section 3.3](#)), as seen in metabolic disorders that affect even the bone health of juveniles. Indeed, as recently proposed, chronic and dysregulated inflammation can be both a product of and contributor to functional declines across many systems and tissues that can generate age-related increases in morbidity and mortality ([Carrasco et al., 2022](#); [Franceschi and Campisi, 2014](#); [McDade, 2023](#)).

3.3. Aging and osteoporosis

Osteoporosis and osteopenia are conditions characterized by decreased bone mineral density, leading to increased susceptibility to fractures. Both are often associated with aging and are more prevalent in elderly individuals, and several studies have explored using bone loss to estimate adult age at death ([Brickley, 2002](#); [Castillo and López Ruiz, 2011](#); [Curate et al., 2022](#); [Gocha et al., 2018](#); [Karydi et al., 2022](#)). However, the mechanisms behind bone loss are complex and studies in a wide variety of populations have emphasized that the idea of a “normative” pattern of bone loss is itself a product of cultural bias (see references in [Agarwal, 2021](#)).

Osteopenia is an early stage of bone loss, preceding osteoporosis, and is diagnosed based on bone mineral density measurements. The molecular mechanisms underlying osteoporosis and osteopenia involve complex interactions between genetic, hormonal, and environmental factors, amplified with aging ([Rachner et al., 2011](#); [Riggs et al., 2004](#); [Taylor et al., 2016](#)). Epigenetic modifications, especially DNA methylation, have been linked to osteoporosis and osteopenia. This is because aging can lead to DNA methylation changes, which can silence or activate the expression of genes involved in bone metabolism ([Bell et al., 2012](#); [Xu et al., 2021](#)). Similarly, hypomethylation of genes that promote osteoclastogenesis can increase bone resorption, further contributing to age-related bone loss ([Seeman and Delmas, 2006](#); [Xu et al., 2021](#)). Research on osteoporosis in ancient populations (e.g., [Agarwal, 2012, 2021](#); [Agarwal and Grynias, 2009](#)) has demonstrated the impact of diet, physical activity post menopause, and childbirth frequency on bone health across time and cultures.

In summary, osteoporosis and osteopenia are linked to aging through hormonal changes, oxidative stress, inflammation, and epigenetic modifications which are themselves linked to culture and behavior. As all of these can also influence the rate of bone remodeling, the effects on microscopic age estimation methods such as morphometry and bone mineral density are important to consider ([Agarwal, 2018](#); [Dominguez and Mavroudas, 2019](#); [Gocha et al., 2018](#); [Hale et al., 2024](#)).

3.4. Osteoarthritis, joint degeneration and age

In 20th–21st century living populations, joint diseases like osteoarthritis (OA) occur at high rates among older adults ([Allen et al., 2022](#)), and are often correlated with aging, although numerous factors contribute to OA ([Anderson and Loeser, 2010](#); [Waldron, 2020](#)). Signs of joint degeneration have sometimes been used to assess age, especially when age-at-death estimation is attempted on highly fragmentary,

commingled, and cremated skeletons ([Strasheim et al., 2023](#); [Winburn and Stock, 2019](#)). However, reliance on the appearance of joint surfaces as a marker of age is complicated by difficulties in differentiating between time-related (i.e., chronological age-related) wear and tear of cartilaginous and bony structures and other inflammatory processes that disrupt surfaces, such as septic arthritis, avascular necrosis, and rheumatoid arthritis, as well as by the relationship between osteoarthritis and activity.

While OA has been associated with higher survivorship in skeletal assemblages from archaeological contexts ([Biehler-Gomez et al., 2025a](#); [Yauussy et al., 2024](#)), the etiology, pathogenesis and clinical form of OA remains a variable composite of genetic, biological, age, and lifestyle factors in living populations ([Strasheim et al., 2023](#); [Waldron, 2019](#)). Despite this, many palaeopathological and bioarchaeological studies pertaining to joint disease address age at death as a covariate ([Baker and Pearson, 2006](#); [Cheverko and Hubbe, 2017](#)), rather than as a dependent variable. More recently, forensic anthropologists have operationalized OA as a potential indicator of age at death alongside more conventional age estimation techniques when constructing biological profiles ([Brenneman et al., 2017](#); [Strasheim et al., 2023](#)). Despite its potential promise in some forensic contexts, the application of OA to age-at-death estimation in paleopathology and bioarchaeology is problematic due to the many interdependent and interactive variables (e.g., genetic predisposition, weight, physical fitness/musculoskeletal strength) that impact the initiation and progression of joint disease. Moreover, manifestations of primary and secondary OA, as well as other arthropathies such as rheumatoid arthritis, share osseous phenotypic markers, but have different causations, ages of onset and disease courses ([Eagle, 2021](#); [Rogers et al., 1987](#)).

3.5. Teeth, oral health and aging

For juveniles, enamel hypoplasia is often used as an indicator of physiological stress during tooth development (e.g., [Goodman et al., 1991](#); [Towle and Irish, 2020](#)), although other factors can also influence their formation and severity ([Cardoso et al., 2024](#); [Lawrence et al., 2021](#); [Taji et al., 2008](#)). It is unclear, however, whether the unknown time span of growth interruption impacts age estimation results based on tooth development stages. Over the life course, pathological changes accumulate in the oral cavity, since the risks of developing dental and oral diseases increase, as dental tissue does not remodel ([Lukacs, 2012](#); [Temple, 2015](#); [Rösing and Kvaal, 1998](#)). Tooth wear plays a key role, especially with respect to developing periapical infections and lesions, and eventual antemortem tooth loss (AMTL) ([Mays, 2014](#); [Morgan, 2012](#)). Much like joint wear, dental wear has been utilized for age estimation (e.g., [Lovejoy, 1985](#); [Miles, 1963](#)), even though the correlations between age-at-death and oral disease prevalence are inherent. Although this relationship is known ([Clarke and Hirsch, 1991](#); [Cucina et al., 2019](#); [Davies-Barrett et al., 2025](#); [Fuchs, 2020](#)), its actual depth and significance is largely overlooked.

Firstly, dental wear removes the crown, starting with the enamel formed early in life and all the direct evidence of health and stress, but also other aspects of lived experience (e.g., diet) archived therein. Consequently, comparing levels of physiological stress derived from simple counts of EH within or between skeletal assemblages from past populations with different mortality structures is not straightforward. Dental wear and antemortem tooth loss (AMTL) should always be considered because it can lead to a loss of information ([Fuchs, 2020](#)). Similarly, advanced caries damage can lead to AMTL. This is why caries correction procedures based on AMTL frequencies, dental preservation, and tooth position susceptibility should also be applied in any study of oral health and age ([Erdal and Duyar, 1999](#); [Lukacs, 1995](#)). Secondly, AMTL is considered to be a typical condition observed among elderly individuals. However, given the wide range of factors influencing oral diseases and eventual AMTL, including genetics, diet, hygiene, occupation and behavior, and socio-economic backgrounds, the

chronological stage when edentulousness is reached differs across periods and contexts of studied populations. Thirdly, oral health is an important determinant of general health. For example, chronic periodontal inflammation, which usually increases with advancing age in industrialized contexts (e.g., Clark et al., 2022), promotes a systemic inflammatory status, which, amongst other things, can affect bone homeostasis, remodeling, and aging (Terkawi et al., 2022).

3.6. Cancer and age

Aging and cancer are closely related (Montégut et al., 2024). Childhood cancer (e.g., leukemia, lymphomas; Siegel et al., 2025; Kaatsch, 2010) have specific age periods when they occur more often, and affected children generally die between 10–14 years (Centers for Disease Control and Prevention, 2015). In adults, the risk of developing malignancies increases during adulthood until it peaks between 85 and 90 years (Siegel et al., 2024). Two main factors responsible for increased frequency of cancers with increasing age include accumulated exposure to carcinogenic impacts and an age-related decrease in the self-regulatory potential of genomic and cellular factors (e.g., DNA repair systems, Niedernhofer et al., 2018). Studies suggest a commonality of hallmarks of aging and cancer, referred to as “agonistic hallmarks” (López-Otín et al., 2023b). These include genomic instability, epigenetic alterations, chronic inflammation, and dysbiosis (Montégut et al., 2024). In contrast, in individuals older than 90 years, some hallmarks of aging seem to protect against the development of malignancies (e.g., telomere attrition and stem cell exhaustion, López-Otín et al., 2023b).

In paleopathology, cancer is considered an age-related disease, but this is difficult to verify in archaeological human remains because identifying neoplastic disease is challenging (Binder et al., 2014; Marques, 2018). Nevertheless, the “paleo-oncological paradigm” traditionally treats cancer as a rare condition in the past (Marques et al., 2021, 2022). This assumption is largely based on the overall lower life expectancy of ancient populations and the effects of evolutionary mismatch over time. Marques and colleagues (2022) challenged this view by adjusting diachronic prevalence rates of malignant neoplasms in ancient skeletal material for both time and age. However, the correlation with age remained insignificant due to uneven representation of the age groups, especially the elderly. In an ideal world, future studies would aim for improved age-adjustment of prevalence rates (Marques et al., 2022), but here again we meet the non-independence of skeletal aging and disease. This may prevent us from finally answering the question of how rare cancer was in past populations.

3.7. Ancient rare diseases and age

Rare diseases form a diverse group of pathological conditions that affect only a small number of people worldwide (Orphanet, 2024). Although there is no single agreed definition for Ancient Rare Diseases (Gresky et al., 2021), eighty percent or more of modern rare diseases are of genetic origin (Orphanet, 2024; Jackson et al., 2018) and, following utilitarian principles, similar prevalences may have also occurred in the past. Most of these Rare Diseases are not usually conceptualized in relation to aging, but many of them have age-related aspects—they can either change the rate of aging or they can change how aging takes place.

Many Ancient Rare Diseases have a predefined specific age of onset. For example, osteopetrosis occurs in two different main types: autosomal-dominant types, which usually start in late childhood/early adulthood and represent the benign form of this skeletal dysplasia (Bollerslev et al., 2013; Gresky et al., 2020), and the autosomal-recessive form, which starts in early life and, without medical intervention, leads to death by age ten (Penna et al., 2021; Nielsen and Alexandersen, 1971). Often, these disease types significantly limit the lifespan of affected individuals. For example, individuals suffering from trisomy 18 die shortly after birth (Goldstein and Nielsen, 1988). Here, a proposed

paleopathological case must match early infant age; otherwise, the diagnosis should be reconsidered (Gresky, 2024). In contrast, individuals born with trisomy 21 before the 20th century could potentially have reached an older age, although the frequently accompanying lung and heart diseases would often have led to an earlier death in childhood (Anastasiadou et al., 2024; Rohrlach et al., 2024) due to the lack of 20th–21st century medical care. The effects of trisomy 21 on the skeleton may include reduced bone density and scoliosis, suggesting that it may affect the appearance of at least some skeletal features.

Very rarely, a disease directly affects biological age by accelerating it. One example is the syndrome complex of progeria (Lessel and Kubisch, 2019), which causes rapid aging from the first year of life onwards until children die around the age of 15 years from a stroke or heart attack (Gordon et al., 2003). However, due to the diverse picture of the syndromes and their non-specific skeletal changes, like small stature, cranio-facial dysplasia (e.g., micrognathia), and osteoporosis, progeria is not easily detectable by macroscopic evaluation in archaeological human remains.

4. Recommendations for age-at-death estimation in paleopathological practice

4.1. Non-independence, bias and data collection

Skeletal age estimation is beset with inherent problems that are well articulated in the field (e.g., Buckberry, 2015; Mays, 2015). The potential for inter- and intra-observer error is also well understood; however, the non-independence of skeletal aging and disease introduces another source of error: visible skeletal lesions of age-related diseases may influence the osteological age-estimation process, producing a systematic bias. This bias is *in addition* to the fact that the same underlying biological processes may affect the appearance of skeletal lesions and the alteration of age estimation markers. Thus, an osteologist who observes markers of age-related disease in the skeleton may consciously or subconsciously adjust the assigned age to reflect this. For example, if using a categorical (i.e., grades, phases) approach to aging, when a marker sits at the boundary between two categories, the presence of osteophytes might prompt the osteologist to put the skeleton into an older category. When repeated for a skeletal series, this would lead to a systematic increase in the apparent age of skeletons with osteophytes in comparison to those without. Hence, the overall difficulty lies in obtaining age information that is as independent of disease as possible while preserving the ability to trace meaningful biological associations between age and disease.

Whilst it seems impossible to correct for the underlying non-independence of skeletal age-estimation markers and pathological lesions, under some circumstances it could be possible to correct for the potential additive bias. One would be to separate out skeletal elements involved in age estimation and then record age-at-death without prior knowledge of pathological conditions: one osteologist/paleopathologist would age all skeletons in the first stage of analysis, and only those skeletal elements used in aging would be observed. Pathologies would then be recorded separately in a later stage of analysis. If two osteologists/paleopathologists were to independently assess age and disease in the same assemblage, the *a priori* bias could be avoided even more readily. In both approaches, the two sets of data could then be combined skeleton by skeleton, resulting in more informed age assessments and evaluations of age-disease related associations. Such protocols would increase the attention towards intra-population variability and characteristics of osseous age-disease relationships.

Generally, standardized data collection that is as comprehensive as possible is likely to be preferable to the prior exclusion of skeletal markers or methods, which would mean a loss of information. Only in this way can possible correlations between age-at-death markers and disease be examined. Case by case, particular markers could be excluded for the estimation of age at death. At the same time, this provides useful

data for future analyses such as in statistical approaches that account for likely covariation of age and disease (e.g., [van Schaik et al., 2018](#)). For instance, whenever a specific subset of diseases is investigated that is clearly age-related, age indicators unaffected by these diseases should be given priority in the assessment of age at death (e.g., research into oral health should not use age-at-death estimation methods based on dental wear, research into osteoarthritis should not be based on the auricular/pubic surface alterations, see [Sections 3.4 and 3.5](#)). Whilst this approach could prove very useful in the assessment of specific diseases, in a general osteological analysis of a skeletal assemblage, this “work around” is not applicable and would rule out multifactorial approaches to aging (e.g., [Clark et al., 2023](#); [Milner et al., 2019](#)).

In an ideal world, the choice of method for estimating the age-at-death of archaeological skeletons would be aligned with their respective in vivo socio-economic, genetic, geographical and cultural backgrounds. Whilst this is not possible in reality, researchers could consider why a particular approach to age-at-death estimation is suitable for the assemblage they are working with and justify that in their work (see [Section 4.2](#)). In any case, it is crucial to explore population-specific morphological variation of skeletal traits, which are critical for both, age-at-death estimations and disease frequencies and distributions.

It is clear that the fundamental problems of bias and methods are not easy to solve, if they can be solved at all. For a practicable approach for the future our best approach may be to explicitly consider such bias. A simple way to achieve this would be to include a brief evaluation of potential biases introduced by age-disease-dependencies as in skeletal reports and studies, especially pathological reports that examine conditions that are associated with aging in the present. Here, possible correlations between specific pathological changes and age markers can be pointed out and discussed, achieving more meaningful information for osteobiographies, highlighting potential confounding factors for palaeoepidemiology, and indicating potential impacts on mortality profiles.

4.2. Adjusting our reporting practice

For decades, researchers have recommended the standardization of age-at-death reporting (see [Buckberry, 2015](#); [Clark et al., 2023](#); [Falys and Lewis, 2011](#)). This usually takes the form either of proposing a standard set of age categories, which come with their own challenges ([Appleby, 2022](#); [Mays, 2015](#); [Roksandic and Armstrong, 2011](#)), or using maximum likelihoods or point estimates of age-at-death and reporting associated errors (e.g., [DeWitte, 2010](#); [Boldsen et al., 2022](#)). Neither approach, however, has led to standardized reporting—skeletal age-at-death estimates are still reported in myriad ways and age categories remain inconsistent. Although the use of standardized age categories offers an apparent ground for comparability of datasets, in reality those categories mean different things in different contexts, since the aging process is contextual and we currently know little about skeletal aging as it pertains to the age indicators conventionally used. Therefore, unless skeletal assemblages are well-matched in genetics, environment, behavior and exposure to variables that influence aging, skeletal age categories will have different relations with chronological age in each population studied. Age categories therefore offer a false idea of comparability. The same is true of point estimates of age-at-death. One must be aware that both the rate and process of skeletal aging can differ between the reference collection used to develop the estimation method and the skeletal assemblage to which it is being applied. As long as we are struggling with this “age mimicry bias” ([Navitainuck et al., 2024](#); [Vossoughi et al., 2022](#); and see [Section 5.3](#)), this is a problem because it seems to rule out our current age-at-death reporting practices. Rather than further entering into this debate, however, here we suggest two steps to integrate into our data-reporting practice which, if widely adopted, could improve the overall usability reproducibility of age data (including for statistical purposes).

First, we argue that it would be enormously helpful to include details

of age-at-death estimation, specifically original scores per skeletal marker and method used, within osteological and paleopathological studies, ideally at both the population and individual level. With this basic morphological information provided, preservation issues, the final age-at-death for each skeleton (or skeletal element), classification into an age category, and possible interactions with pathological processes are visible to every researcher. This enables us to consider population-specific aging processes and age structures in future projects that make use of existing raw data, which is in turn useful for life history, frailty and modeling approaches. These data do not need to be presented or discussed in detail, but could be made available in supplementary materials or in freely accessible repositories (e.g., digital tables). Such reporting would be not only straightforward and quick to implement, since these data are always generated during osteological analysis, but would also comply with FAIR principles for data management ([Wilkinson et al., 2016](#)). Although we are primarily concerned with age, this recommendation could also extend to other aspects of osteological and paleopathological work, such as sex estimation.

Secondly, whenever possible, but specifically in case reports and osteobiographies, it makes sense to include individual-level considerations of potential age-disease interactions. In studies of single assemblages, researchers could explicitly consider potential interactions of aging and disease at the population-level and any relevant sub-levels (e.g., sex, socioeconomic status). In multi-assemblage studies, researchers could consider likely differences in age-disease interactions between populations and the potential implications of these for the interpretation of demographic differences in disease and frailty.

These two steps may seem trivial at first, but they could have a transformative effect when applied widely and in the long term. They would enable us as a field to move away from the all-too-common situation wherein the relationship between aging and disease sits in an unexamined black box, instead foregrounding this relationship as an overt focus of interest and research. It is likely that this will enable enhanced understanding both of aging and of disease in the past.

5. Recommendations for future research

5.1. Physiological mechanisms of skeletal age markers

On a practical level, there has been little dedicated research into the mechanisms driving changes in skeletal age markers over the life course. Our information is largely extrapolated from medical, physiological, and anatomical studies in which skeletal aging was not the main focus (e.g., [Mays, 2015](#)). Whilst this gives us a certain amount of foundational knowledge, we remain ignorant of the specific interaction of disease processes with remodeling of skeletal features used in age-at-death estimation. Taking inspiration from the pathophysiological approach in paleopathology (e.g., [Mays, 2015](#)), there is room to consider potential mechanisms of disease and their likely effects on skeletal age markers. The relationship between chronic, low-level inflammation and skeletal aging, for instance, is poised for such research (see [Section 3.2](#)).

In order to study the interaction between osseous and soft tissues, radiological data (CT, x-ray) from forensic cases and living populations can be used. Written records of premodern or forensic skeletal collections do not deliver the same level of information, yet might be helpful in answering specific case-related questions. Thus, in practice, research into the relationship between disease and skeletal age markers could be developed through the use of modern clinical data ([Anderson et al., 2025](#); [van Schaik et al., 2014](#); [2018](#)), and recent to modern skeletal collections for which chronological ages and health data are available, although resolution and capture of surface features remain an issue. Where modern clinical data is used, appropriate attention will also need to be paid to relevant ethical principles (e.g., [CARE Principles, 2025](#)).

Imaging studies in living patients have great potential, but may pose significant access restrictions that limit their application ([Stull et al., 2021](#)). Large amounts of data are also being accrued in forensic

pathology through postmortem CT scanning, which, in contrast to that of living patients, provide opportunities to recognize changes in dry bone morphologies. Combined with autopsy findings and health data, as well as modern diagnostics and treatments, this may be translatable to paleopathological research. Where we have documented reference skeletal collections with known medical histories (e.g., donated collections), we can develop case-control studies exploring differences in age markers between individuals with and without a specific diagnosed disease (see Bertsatos et al., 2020; Milner and Boldsen, 2017). Obtaining sufficient sample sizes that control for factors other than the disease of interest will be a key obstacle to overcome (for successful attempts see Daneshvari and Edgar, 2021; Joshi and Tallman, 2023; Wyatt and O'Donnell, 2025). Where sequential scans are available for living individuals, we can carry out case-control studies to investigate how skeletal age markers change over time and how they relate to disease history, potentially alongside genetic and epigenetic profiles. Finding suitable and large enough samples is likely to be difficult, although the increasing availability of digital datasets may help with this. One challenge in extrapolating from clinical data for paleopathology is the non-comparability of modern and past populations because of differing biosocial and environmental conditions and human plasticity, amongst other variables. This is clearly evident for aspects of nutrition and physical activity that directly influence bone remodeling and morphology, as well as for medical and therapeutic measures.

For developing age-at-death estimation methods, there is a need for investigations of more phenotypically, genetically, geographically and socially diverse skeletal collections (for example, as pioneered in the development of the transition analysis method (Boldsen et al., 2002; Milner et al., 2019)). Since a close correspondence between reference and archaeological skeletal collections is lacking, inclusion of varied populations when testing and validating new age-at-death estimation methods will enable us to investigate the effects of different environmental and biosocial conditions on the morphology of age markers. In turn, this will allow researchers to consider these parameters when choosing age-at-death estimation methods and selecting methods most appropriate for the assemblage they are analysing.

New techniques, such as CT scanning and radiometrics, can also help visualize age-informative bone structures and changes within known-age individuals. Scanning, alongside machine-learning approaches using quantitative use of 3D data, may help us understand shape and textural changes and may reduce the effect of intra- and interobserver bias (Barszcz and Woźniak, 2025).

5.2. Biomedical prospects

There are promising biomedical perspectives to better understand the morphology of skeletal markers and skeletal age-disease relationships. As proposed for the potential relationship between physiological stress and age-at-death estimations (Simon et al., 2023; Raghu et al., 2021), we suggest that paleopathological studies explicitly consider the interplay of inflammatory phenotypes and biological age (see Section 3.2). For example, we could consider bone alterations, such as those associated with periodontal disease, as markers of chronic systemic inflammation and use these to reconstruct skeletal inflammatory phenotypes for past individuals and populations (Crespo, 2020). From an evolutionary perspective, we may be able to integrate recent paleogenetic research on changes in immune genes, cytokine production capacities, and the development of autoimmune diseases (e.g., da Silva et al., 2025; Domínguez-Andrés et al. 2021, Krause-Kyora et al., 2025; Plascencia et al., 2025) into paleopathological research. Paleopathology would benefit from considering potential skeletal, oral, biomolecular and genetic markers of inflammaging and related processes of chronic physiological stress (e.g., Forshaw, 2024; Webb et al., 2015), as well as systemic inflammatory phenotypes, in their assessments of disease relative to age, aging, and even age-at-death estimation.

Paleopathology will also benefit from new approaches such as

epigenetics and metabolomics in order to better understand processes of skeletal aging and disease. For example, there is increasing evidence that DNA methylation plays a role in aging processes and contributes significantly to inflammaging (Wang et al., 2022). Indeed, the degree of DNA methylation is a biomarker of aging associated with *age-related disease* and all-cause mortality. Metabolomic clocks also offer information about the aging process (see Huang et al., 2025). Current research has established that bone is a suitable target for metabolomic research (Badillo-Sanchez et al., 2023; Bonicelli et al., 2024). Epigenetics and omics-based approaches, therefore, hold potential to help us to better understand both skeletal aging, past *age-related disease* and *disease-related age*. While these methods have the potential to include samples from chronologically older, fragmentary, and commingled contexts, they are destructive, relatively expensive, and require excellent preservation of skeletal tissue.

5.3. Statistics, modeling and “big data”

While individual life histories often reveal potential interconnections between aging and disease, potential also lies in statistical analysis and modeling. *Age-related disease* and *disease-related age* interactions could be incorporated into Bayesian models of aging and age-at-death estimations. For instance, the use of priors in Bayesian statistical approaches could be used in advanced probabilistic modeling approaches for testing relationships between parameters of aging (e.g., diet, occupation, genetics) and disease (e.g., physical strain, pathogen exposure) at different levels of analysis from the skeletal marker, to individual-level-age profiles, to cross-population studies. The approaches could also help address the issue of age mimicry bias in our traditional methods (see Navitainuck et al., 2024; Vossoughi et al., 2022). The latest paleodemographic research has demonstrated its functional power (e.g., Hens and Godde, 2022; Müller-Scheeßel et al., 2025), operable by multiple sources of information and with different data qualities (i.e., fragmentary material). With machine learning and artificial intelligence, big data approaches also offer possibilities for pattern recognition. However, this requires sufficient standardized osteological and paleopathological data. Existing data repositories, such as the one generated by the Global History of Health project (Steckel et al., 2018), could potentially be used for basic tests of age-disease interrelations—“basic” because the original age marker scores are not (yet) available (see Section 4.2).

Techniques such as TA3 (Milner et al., 2019) take a holistic approach to offer more accurate age-at-death estimations across populations (although validation studies have yet to be undertaken on skeletal collections different from those used when developing the method). Keeping in mind that age estimates are affected by the relationship between age and disease, the inclusion in the TA3 reference sample of individuals, regardless of their life histories, renders confidence intervals longer than if we could adequately control for life (including disease) variables. As a large number of skeletal markers are included in the TA3 assessment the technique appears promising for disparate skeletal assemblages and, even for commingled and disarticulated skeletal remains (Bolster et al., 2024; Getz, 2020). Further research into the mechanisms underlying the skeletal features included within TA3 might allow refinement of the method and an associated reduction in confidence intervals.

6. Conclusion

The aim of this perspective paper was to draw attention to a complicated and ubiquitous, yet largely unaddressed, topic within osteology, paleopathology and paleoepidemiology: the relationship between skeletal age and disease. We outline a framework to aid in recognizing the effects of age-disease interrelations and discuss the significance of age-related disease and disease-related age using paleopathological examples. These examples demonstrate the multifaceted nature of the relationships between disease and aging. Additionally, we

make clear that the osteological standard parameter “age at death” is not an isolated variable, since skeletal age markers impact disease processes and disease markers affect aging. We argue success in understanding health and disease in the past likely lies in promoting interdisciplinary and transdisciplinary exchange between osteologists, paleopathologists, clinicians, radiologists, immunologists, geneticists, gerontologists, human biologists, bioarchaeologists, paleodemographers and archeologists. We hope that this paper marks a first step in this direction.

CRedit authorship contribution statement

Kathryn E. Marklein: Writing – review & editing, Writing – original draft, Investigation. **Molly Zuckerman:** Writing – review & editing, Writing – original draft, Investigation. **Marie Louise Schjellerup Jørkov:** Writing – review & editing, Investigation, Conceptualization. **George R. Milner:** Writing – review & editing, Writing – original draft, Investigation. **Katharina Fuchs:** Writing – review & editing, Writing – original draft, Project administration, Investigation, Funding acquisition, Conceptualization. **Jo Appleby:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Julia Gresky:** Writing – review & editing, Writing – original draft, Investigation. **Fabian Crespo:** Writing – review & editing, Writing – original draft, Investigation. **Niels Lynnerup:** Writing – review & editing, Writing – original draft, Investigation. **Katherine van Schaik:** Writing – review & editing, Writing – original draft, Investigation.

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